



UNIVERSITEIT VAN AMSTERDAM

Mathematical Biology

NOTES BY

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INTRODUCTION. In the course, much attention is paid to “translation”: how do we get from biological information to a mathematical formulation of questions? And what do the mathematical results tell us about biological phenomena? In addition, the course aims to introduce general physical ideas about time scales and spatial scales and how these can be used to great advantage when performing a mathematical analysis.

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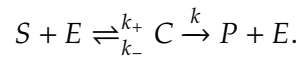
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Exploiting Time Scale Differences: the Quasi-Steady-State Approximation (QSSA)

SECTION 1.1

| Michaelis-Menten Enzyme Kinetics

Enzymes are large protein molecules that catalyse reactions in the living cell. The simplest situation is that they transform substrate into product (by way of a conformational change) according to the reaction scheme



SECTION 1.2

| Scaling

SECTION 1.3

| Excitability

Phase Plane Analysis of Prey-Predator Systems

SECTION 2.1

| The Story of d'Ancona and Volterra

During World War I, Italian fishermen were forced to keep their boats ashore along the Adriatic coast because of the danger of being sunk. With relief, they resumed fishing after the war ended. Understandably, it vexed them to discover that the percentage of shark-like predatory fish (which did not fetch an attractive price) in the haul had increased considerably compared with the pre-war period.

Hearing their complaints, the biologist d'Ancona wondered about an explanation. Unable to produce one, he posed the puzzle to his father-in-law, a famous mathematician named Volterra, who produced, first of all, the system of two differential equations

$$\frac{dv}{dt} = av - bvp, \quad (2.1)$$

$$\frac{dp}{dt} = dvp - cp. \quad (2.2)$$

Here, v stands for victim (the prey) and p for predator, and a, b, c and d are non-negative parameters.

EXERCISE 2.1. Explain the modelling assumptions underlying this system. In particular, explain the assumptions underlying the form of the terms.

SOLUTION. This model assumes the existence of two interacting populations in the ecosystem: v for victims and p for predators, with a, b, c, d being non-negative parameters.

- The rate of change of the prey population v is influenced by:
 - Growth of prey av : It represents the intrinsic growth rate of the prey population in the absence of predators.
 - Predation bvp : It represents the rate at which predators consume prey. b represents the predation rate, and it assumes that the rate of predation is directly proportional to the population size of prey and predators.
- The rate of change of the predator population p is influenced by:
 - Death of predator cp : It represents the natural death rate of predators in the absence of prey.

- Successful predation dvp : It represents the rate at which predators successfully capture and consume prey. It assumes that the rate of predation is proportional to the population size of both predators and prey. d represents the success rate of predation.

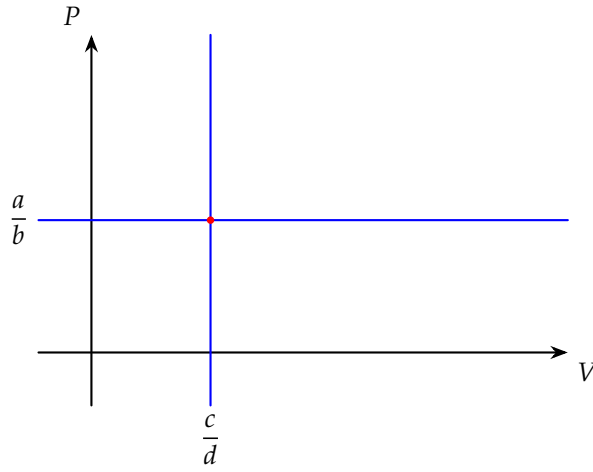
These equations describe how populations change continuously under simplified assumptions about interactions between prey and predators. The model does not consider factors such as competition among predators or resource limits. $\square$

EXERCISE 2.2. Draw the null-clines (i.e., zero isoclines). Indicate the steady states by dots.

SOLUTION.

$$\frac{dv}{dt} = av - bvp = 0 \implies p = \frac{a}{b}, \quad (2.3)$$

$$\frac{dp}{dt} = dvp - cp = 0 \implies v = \frac{c}{d}. \quad (2.4)$$



$\square$

EXERCISE 2.3. Express the non-trivial steady state in terms of the parameters a, b, c and d .

SOLUTION. From (2.3)–(2.4), if $v \neq 0$, then $a - bp = 0$, which implies $p = \frac{a}{b}$; if $p \neq 0$, then $dvp - c = 0$, which implies $v = \frac{c}{d}$. $\square$

EXERCISE 2.4. The effect of fishing can be captured by replacing a by $a - \mu$ and $-c$ by $-c - \mu$. Explain the rationale underlying this statement.

SOLUTION. μ represents the per capita fishing mortality rate. Fishing therefore reduces the prey's intrinsic growth rate from a to $a - \mu$ and increases the predator's mortality rate from c to $c + \mu$, giving the term $-(c + \mu)p$. $\square$

EXERCISE 2.5. Reproduce Volterra's explanation of the change in predator and prey fish before and after World War I.

SOLUTION. Before the war, normal fishing activities affected the natural predator-prey dynamics. When fishing resumed after the war, there was a notable change in catch composition: the percentage of less economically valuable, shark-like predatory fish had increased. Fishing reduces both populations proportionally, which lowers their average equilibrium densities; because the predator equilibrium is proportional to the prey growth rate whereas the prey equilibrium is proportional to predator mortality, the relative effect explains the observed change in the catch. $\square$

EXERCISE 2.6. When spraying insecticides to protect a crop from herbivorous insects, one should verify first whether the herbivores are currently kept in check (perhaps, from the farmer's point of view, at an unacceptably high density) by a natural enemy which is sensitive to the insecticide too. Why?

SOLUTION. (i) Preservation of natural predators: Some natural enemies play an important role in keeping pest populations in check. If we do not account for this, the insecticide may harm these beneficial species.

(ii) Ecological balance: Natural enemies help prevent pest outbreaks and reduce the need for chemical insecticides. Disrupting this balance can lead to potentially worse problems.

(iii) Reduced pesticide dependence: Pesticides may have adverse effects on other non-target organisms and lead to unintended consequences.

$\square$

SECTION 2.2

| The Phase Portrait of the Volterra-Lotka System

The zero-isoclines divide the positive quarter of the (v, p) -plane into four regions.

EXERCISE 2.7. Give a schematic picture of the direction of the flow by putting arrows on the zero-isoclines and in each of the four regions (in the spirit of wind directions: north-east, etc.).

SOLUTION. □

At a glance we see that orbits have a tendency to spin around the non-trivial steady state, but we can't see whether they spiral inward or outward. In an attempt to determine this very close to the steady state, we use linearisation.

EXERCISE 2.8. Compute the Jacobian matrix corresponding to the non-trivial steady state. Compute the eigenvalues of this matrix. What do you conclude?

SOLUTION.

$$\frac{dv}{dt} = f(v, p), \quad \frac{dp}{dt} = g(v, p).$$

For $\frac{dv}{dt} = 0$,

$$\frac{\partial f}{\partial v} = a - bp, \quad \frac{\partial f}{\partial p} = -bv.$$

For $\frac{dp}{dt} = 0$,

$$\frac{\partial g}{\partial v} = dp, \quad \frac{\partial g}{\partial p} = dv - c.$$

Then

$$J = \begin{bmatrix} \frac{\partial f}{\partial v} & \frac{\partial f}{\partial p} \\ \frac{\partial g}{\partial v} & \frac{\partial g}{\partial p} \end{bmatrix} = \begin{bmatrix} a - bp & -bv \\ dp & dv - c \end{bmatrix}$$

$\det(J - \lambda I) = 0$, non-trivial steady state $(\frac{c}{d}, \frac{a}{b})$. Therefore,

$$J(v^*, p^*) = \begin{bmatrix} a - b\frac{a}{b} & -b\frac{c}{d} \\ d\frac{a}{b} & d\frac{c}{d} - c \end{bmatrix} = \begin{bmatrix} 0 & -\frac{bc}{d} \\ \frac{ad}{b} & 0 \end{bmatrix}.$$

$$\det(J - \lambda I) = \begin{vmatrix} -\lambda & -\frac{bc}{d} \\ \frac{ad}{b} & -\lambda \end{vmatrix} = \lambda^2 + \frac{abcd}{bd} = 0.$$

Hence $\lambda^2 = -ac$, which implies $\lambda_{1,2} = \pm i\sqrt{ac}$. □

In fact, the Volterra-Lotka¹ phase portrait is rather special: it consists of a collection of nested closed orbits. To demonstrate this, the function $L : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ defined by

$$L(v, p) = dv - c \log v + bp - a \log p \tag{2.5}$$

¹ After the American mathematical biologist Lotka, who formulated exactly the same system of differential equations completely independently at about the same time.

is most helpful. The point is that L is a *constant of motion* (also called a *conserved quantity*), which means that the value of L does not change along an orbit of (2.1)–(2.2). In other words: orbits are contained in level sets of L .

We have to do three things:

- (1) verify that indeed L is a constant of motion;
- (2) explain how one derives L from (2.1)–(2.2) (this in order to explain how one could come to the idea of studying L in relation to (2.1)–(2.2));
- (3) use L to show that the orbits of (2.1)–(2.2) are closed curves.

EXERCISE 2.9. Compute

$$\frac{d}{dt}L(v(t), p(t))$$

for an arbitrary solution $t \mapsto (v(t), p(t))$ of (2.1)–(2.2). Does it follow that L is a constant of motion?

SOLUTION. By the chain rule,

$$\frac{d}{dt} \log v = \frac{1}{v} \cdot \frac{dv}{dt}, \quad \frac{d}{dt} \log p = \frac{1}{p} \cdot \frac{dp}{dt}.$$

Then

$$\begin{aligned} \frac{dL}{dt} &= \frac{\partial L}{\partial v} \cdot \frac{dv}{dt} + \frac{\partial L}{\partial p} \cdot \frac{dp}{dt} \\ &= \left(d - \frac{c}{v}\right)(av - bvp) + \left(b - \frac{a}{p}\right)(dvp - cp) \\ &= adv - bdpv - ac + bcp + bdpv - bcp - adv + ac = 0. \end{aligned}$$

□

EXERCISE 2.10. With (2.1)–(2.2) as a starting point, we compute

$$\frac{dp}{dv} = \frac{\frac{dp}{dt}}{\frac{dv}{dt}} = \frac{(-c + dv)p}{(a - bp)v} = \frac{-\frac{c}{v} + d}{\frac{a}{p} - b},$$

(while simply ignoring that the denominator does occasionally become zero). Use *separation of variables* to solve this differential equation. What do you conclude?

SOLUTION. We need to solve

$$\int \left(\frac{a}{p} - b\right) dp = \int \left(-\frac{c}{v} + d\right) dv,$$

which is equivalent to

$$a \log p - bp + C_1 = -c \log v + dv + C_2,$$

i.e.,

$$a \log p - bp = -c \log v + dv + C,$$

where C is a constant. Rearranging gives

$$dv - c \log v + bp - a \log p = C',$$

so $L(v, p)$ is constant along trajectories.

□

EXERCISE 2.11. Draw the graph of the function $v \mapsto dv - c \log v$. Conclude that the equation

$$dv - c \log v - c + c \log \frac{c}{d} = \varepsilon$$

has, for given $\varepsilon > 0$, exactly two solutions $v_{\pm}(\varepsilon)$. Repeat this analysis for $p \mapsto bp - a \log p$. Next, consider for given $\theta > 0$ the equation

$$L(v, p) - L\left(\frac{c}{d}, \frac{a}{b}\right) = \theta.$$

Check that, for any ε with $0 < \varepsilon < \theta$ the four points $(v_{\pm}(\varepsilon), p_{\pm}(\theta - \varepsilon))$ are solutions. Thus, we obtain four paths, parametrised by ε , on which L takes the constant value $L\left(\frac{c}{d}, \frac{a}{b}\right) + \theta$. Check that, by inclusion of the limiting points for $\varepsilon \rightarrow 0$ and $\varepsilon \rightarrow \theta$, we obtain a closed curve.

SOLUTION. Let

$$L(v, p) = dv - c \log v + bp - a \log p, \quad v > 0, p > 0,$$

and we write

$$\begin{aligned} f(v) &= dv - c \log v - c + c \log \frac{c}{d}, \\ g(p) &= bp - a \log p - a + a \log \frac{a}{b}. \end{aligned}$$

Then

$$\begin{aligned} f'(v) &= d - \frac{c}{v} = 0 \implies v = \frac{c}{d}, \\ g'(p) &= b - \frac{a}{p} = 0 \implies p = \frac{a}{b}. \end{aligned}$$

Together with $f''(v) = \frac{c}{v^2} > 0$ and $g''(p) = \frac{a}{p^2} > 0$, $f(v)$ is strictly convex on $(0, \infty)$ with a unique critical point $v = \frac{c}{d}$, and at this point

$$f\left(\frac{c}{d}\right) = \frac{cd}{d} - c \log \frac{c}{d} - c + c \log \frac{c}{d} = 0.$$

Moreover,

$$\lim_{v \rightarrow 0^+} f(v) = \infty, \quad \lim_{v \rightarrow \infty} f(v) = \infty.$$

Therefore, for every $\varepsilon > 0$ the equation $f(v) = \varepsilon$ has exactly two solutions, and the same argument works for $g(p)$.

$$L(v, p) - L\left(\frac{c}{d}, \frac{a}{b}\right) = \theta,$$

and, by the definitions above, this is simply

$$f(v) + g(p) = \theta.$$

Thus, for each $0 < \varepsilon < \theta$, the four points $(v_{\pm}(\varepsilon), p_{\pm}(\theta - \varepsilon))$ lie on the same level set. $\square$

So the Volterra–Lotka model predicts persistent prey–predator oscillations with an amplitude determined by the choice of initial conditions. Yet, in *Section 3.1*, Volterra’s answer to d’Ancona was based on steady-state values. Is there not a discrepancy?

EXERCISE 2.12. Show that the average values of v and p over one period are equal to the steady state values, i.e.,

$$\frac{1}{T} \int_0^T v(\tau) d\tau = \frac{c}{d}, \quad \frac{1}{T} \int_0^T p(\tau) d\tau = \frac{a}{b},$$

where T is the period (corresponding to one full turn of (v, p) along a closed orbit).

Hint: divide the first equation of (2.1)–(2.2) by v and the second by p ; use that $\frac{d}{dt} \log v(t) = \frac{1}{v(t)} \frac{dv}{dt}(t)$ and that $v(T) = v(0)$, together with the analogous identities for p .

SOLUTION.

$$\begin{aligned} \begin{cases} \frac{dv}{dt} = av - bvp \\ \frac{dp}{dt} = dvp - cp \end{cases} &\implies \begin{cases} \frac{1}{v} \frac{dv}{dt} = a - bp \\ \frac{1}{p} \frac{dp}{dt} = dv - c \end{cases} \implies \begin{cases} \frac{d \ln v(t)}{dt} = a - bp \\ \frac{d \ln p(t)}{dt} = dv - c \end{cases} . \\ \int_0^T \frac{d \ln v(t)}{dt} dt &= \int_0^T (a - bp) dt \\ \implies \ln v(t) \Big|_0^T &= 0 \\ \implies \int_0^T (a - bp) dt &= 0 \\ \implies \int_0^T a dt - b \int_0^T p dt &= 0 \\ \implies aT - b \int_0^T p dt = 0 &\implies \frac{1}{T} \int_0^T p dt = \frac{a}{b}. \end{aligned}$$

For the same reason:

$$\frac{1}{T} \int_0^T v(t) dt = \frac{c}{d}.$$

□

We conclude with two side remarks, formulated as exercises.

EXERCISE 2.13. Show that the trivial steady state is a saddle point.

SOLUTION.

$$\begin{cases} \frac{dv}{dt} = av - bvp \\ \frac{dp}{dt} = dvp - cp \end{cases} \implies \begin{cases} 0 = av_0 - bv_0p_0 \\ 0 = dv_0p_0 - cp_0 \end{cases} \implies v_0 = 0, p_0 = 0.$$

Trivial steady state is $(0, 0)$.

$$J = \begin{bmatrix} a - bp & -bv \\ dp & dv - c \end{bmatrix} \implies J(0, 0) = \begin{bmatrix} a & 0 \\ 0 & -c \end{bmatrix}.$$

Let $\det(J(0, 0) - \lambda I) = 0$,

$$\det \begin{bmatrix} a - \lambda & 0 \\ 0 & -c - \lambda \end{bmatrix} = (a - \lambda)(-c - \lambda) = 0$$

We have $\lambda_1 = a$, $\lambda_2 = -c$. Therefore $(0, 0)$ is a saddle point.

□

EXERCISE 2.14. Show that the number of parameters in (2.1)–(2.2) can be reduced from 4 to 1 by scaling of v , p and t .

SOLUTION.

$$\begin{aligned} \frac{dv}{dt} &= av - bvp \\ \frac{dp}{dt} &= dvp - cp \end{aligned}$$

Choose $v' = \frac{d}{c}v$, $p' = \frac{b}{a}p$, $t' = at$.

$$\begin{cases} \frac{dv'}{dt'} = v' - v'p' \\ \frac{dp'}{dt'} = \lambda(v'p' - p') \end{cases} \quad (\lambda = \frac{c}{a})$$

$$\begin{aligned} \begin{cases} \frac{dv'}{dt'} \frac{dv}{dt} \frac{dt}{dt'} = \frac{d}{c}v - \frac{d}{c}v \frac{b}{a}p \\ \frac{dp'}{dt'} \frac{dp}{dt} \frac{dt}{dt'} = \frac{c}{a} \left(\frac{d}{c}v \frac{b}{a}p - \frac{b}{a}p \right) \end{cases} &\implies \begin{cases} \frac{d}{c} \frac{dv}{dt} \frac{1}{a} = \frac{d}{c}v - \frac{bd}{ac}pv \\ \frac{b}{a} \frac{dp}{dt} \frac{1}{a} = \frac{bd}{a^2}vp - \frac{bc}{a^2}p \end{cases} \\ &\implies \begin{cases} \frac{dv}{dt} = \frac{dv}{c} \frac{ac}{d} - \frac{bd}{ac} \frac{ac}{d}vp \\ \frac{dp}{dt} = \frac{bd}{a^2} \frac{a^2}{b} - \frac{bc}{a^2} \frac{a^2}{b}p \end{cases} \\ &\implies \begin{cases} \frac{dv}{dt} = av - bvp \\ \frac{dp}{dt} = dvp - cp \end{cases} \end{aligned}$$

So letting $\lambda = \frac{c}{a}$ we obtain

$$\begin{cases} \frac{dv'}{dt'} = v' - v'p' \\ \frac{dp'}{dt'} = \lambda(v'p' - p') \end{cases}$$

□

SECTION 2.3

| The Effect of Limitations in Prey Growth

The Volterra–Lotka model gives an oversimplified description of prey–predator interaction. Yet it is a convenient reference point for the incorporation of additional mechanisms. In particular, the structural instability, as exemplified by the neutral stability of the steady state and the family of periodic orbits, allows us to classify such mechanisms as either *stabilising* or *destabilising*, depending on whether the steady state in the modified model is (locally asymptotically) stable or unstable. In this and some of the following sections, we discuss various modifications in this spirit. Meanwhile, we enlarge our arsenal of phase-plane techniques and practise their use.

The so-called logistic equation

$$\frac{dv}{dt} = av - ev^2 = av\left(1 - \frac{e}{a}v\right) \quad (2.6)$$

describes, in a phenomenological manner, that, even in the absence of a predator, prey growth may be limited, with prey population size settling down at the so-called *carrying capacity*

$$\tilde{v} = \frac{a}{e}. \quad (2.7)$$

The non-negative parameter e describes the detrimental effects of crowding.

When $-c + d\tilde{v} < 0$, the predator population declines even at the maximal sustainable level of prey. We expect that the predator is doomed to go extinct.

EXERCISE 2.15. Determine the phase portrait of the system

$$\frac{dv}{dt} = v(a - ev - bp), \quad (2.8)$$

$$\frac{dp}{dt} = p(-c + dv), \quad (2.9)$$

for parameter combinations such that

$$-c + d\frac{a}{e} < 0. \quad (2.10)$$

In particular, draw the isoclines and a “wind-direction” vector field. Show by geometric arguments that the steady state $(\frac{a}{e}, 0)$ is globally asymptotically stable. Interpret this conclusion biologically.

SOLUTION. Let $\frac{dv}{dt} = 0$.

$$v(a - ev - bp) = 0$$

Then $v = 0$, or $a - ev - bp = 0$ and then $p = \frac{a-ev}{b}$. Let $\frac{dp}{dt} = 0$.

$$p(-c + dv) = 0$$

Then $p = 0$, or $-c + dv = 0$ and hence $v = \frac{c}{d}$. By (2.10), $\frac{dp}{dt} < 0$ when $v = \frac{a}{e}$.

Visualising this in the phase plane, the nullclines divide the plane into regions, and the vector fields direct all trajectories towards the steady state $(\frac{a}{e}, 0)$. Hence, no matter where you start in the first quadrant, the system will evolve towards this steady state. This makes $(\frac{a}{e}, 0)$ globally asymptotically stable in the first quadrant.

It means that regardless of the initial predator p and prey v numbers, the predator population will eventually become extinct, and the prey population will be stable at its carrying capacity. Even if the predator can consume prey, the prey population's intrinsic growth dynamics and the given constraints can still lead to the predator's extinction. $\square$

When, on the other hand, we have the opposite inequality,

$$-c + d\frac{a}{e} > 0, \quad (2.11)$$

the predator population will start growing when introduced in an environment in which the prey is at its carrying capacity. We say that the predator can invade successfully.

EXERCISE 2.16. Assume (2.11). Show that

- (i) $(\frac{a}{e}, 0)$ is a saddle point;
- (ii) the non-trivial steady state

$$\left(\frac{c}{d}, \frac{ad - ec}{bd} \right)$$

is biologically meaningful, and that it is either a stable node or a stable spiral.

Next, draw the isoclines and the wind-direction field for this case.

SOLUTION. (i)

$$\frac{dv}{dt} = v(a - ev - bp), \quad \frac{dp}{dt} = p(-c + dv).$$

$$J = \begin{bmatrix} a - 2ev - bp & -bv \\ dp & -c + dv \end{bmatrix}.$$

$$J\left(\frac{a}{e}, 0\right) = \begin{bmatrix} -a & -\frac{ab}{e} \\ 0 & -c + \frac{ad}{e} \end{bmatrix}.$$

$$\det(J - \lambda I) = \det \begin{bmatrix} -a - \lambda & -\frac{ab}{e} \\ 0 & \frac{ad}{e} - c - \lambda \end{bmatrix} = (-a - \lambda) \left(\frac{ad}{e} - c - \lambda \right).$$

$$\lambda_1 = -a, \quad \lambda_2 = \frac{ad}{e} - c.$$

One eigenvalue is positive and the other is negative; hence $(\frac{a}{e}, 0)$ is a saddle point.

(ii)

$$J\left(\frac{c}{d}, \frac{ad - ec}{bd}\right) = \begin{bmatrix} a - 2e\frac{c}{d} - b\frac{ad - ec}{bd} & -b\frac{c}{d} \\ d\frac{ad - ec}{bd} & -c + d\frac{c}{d} \end{bmatrix} = \begin{bmatrix} a - \frac{2ec}{d} & -\frac{bc}{d} \\ \frac{ad - ec}{b} & 0 \end{bmatrix}$$

$$\det(J - \lambda I) = \det \begin{bmatrix} a - \frac{2ec}{d} - \lambda & -\frac{bc}{d} \\ \frac{ad - ec}{b} & -\lambda \end{bmatrix} = \left(\lambda + \frac{ec}{d} \right) \lambda + \frac{bc}{d} \frac{ad - ec}{b} = 0$$

$$\lambda = \frac{-\frac{ec}{d} \pm \sqrt{\left(\frac{ec}{d}\right)^2 - \frac{4bc}{d} \frac{ad - ec}{b}}}{2}$$

$$-\frac{ec}{d} + \sqrt{\left(\frac{ec}{d}\right)^2 - 4c \frac{ad - ec}{d}} \leq \frac{-ec}{d} + \sqrt{\left(\frac{ec}{d}\right)^2} - \sqrt{4c \frac{ad - ec}{d}} = \sqrt{-4c \frac{ad - ec}{d}}.$$

Thus, when $\Delta = \left(\frac{ec}{d}\right)^2 - 4c \frac{ad - ec}{d} > 0$, the eigenvalues are two negative real numbers and the fixed point is stable. When $\Delta < 0$, the eigenvalues are complex with negative real parts, and the fixed point is a stable spiral. $\square$

In fact, the non-trivial steady state is *globally* asymptotically stable when (2.11) holds. In other words, orbits starting away from the steady state spiral in towards it. To demonstrate this, we use an auxiliary function², which is decreasing along orbits, and which achieves its minimum in the non-trivial steady state. Such functions are called *Lyapunov functions* and often denoted by the symbol V . There is no general method for their construction.³ One has to rely on luck, intuition, experience and perseverance.

We define $V : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ by

$$V(v, p) = d \left(v - \frac{c}{d} \log v \right) + b \left(p - \frac{ad - ec}{bd} \log p \right). \quad (2.12)$$

EXERCISE 2.17. Show that V decreases along orbits of (2.9). More precisely, show that

$$\frac{d}{dt} V(v(t), p(t)) \leq 0$$

for an arbitrary solution $t \mapsto (v(t), p(t))$ of (2.9), with equality if and only if $v(t) = \frac{c}{d}$.

SOLUTION.

$$\frac{dV}{dt} = \frac{\partial V}{\partial v} \frac{dv}{dt} + \frac{\partial V}{\partial p} \frac{dp}{dt}, \quad \frac{\partial V}{\partial v} = d - \frac{c}{v}, \quad \frac{\partial V}{\partial p} = b \left(1 - \frac{ad - ec}{bdp} \right) = b - \frac{ad - ec}{dp}.$$

Therefore

$$\begin{aligned} \frac{dV}{dt} &= \left(d - \frac{c}{v} \right) v(a - ev - bp) + \left(b - \frac{ad - ec}{dp} \right) p(-c + dv) \\ &= (dv - c)(a - ev - bp) + \left(bp - \frac{ad - ec}{d} \right) (-c + dv) \\ &= (-c + dv) \left(a - ev - bp + bp - \frac{ad - ec}{d} \right) \\ &= (-c + dv) \left(-ev + \frac{ec}{d} \right) \\ &= evc - \frac{ec^2}{d} - edv^2 + ecv \\ &= e \left(2vc - \frac{c^2}{d} - dv^2 \right) = -e \left(\sqrt{d}v - \frac{c}{\sqrt{d}} \right)^2. \end{aligned}$$

Let $\frac{dV}{dt} = 0$, then $v = \frac{c}{d}$. □

From this information, one can deduce that every orbit approaches the non-trivial steady state. The precise argument is a bit technical but goes roughly like this: points in the ω -limit set⁴ of an arbitrary orbit must lie in the same level set of V and all orbits starting in such ω -limit points must lie in that same level set too; only the non-trivial steady state satisfies these requirements.⁵

² A slight modification of L introduced in the previous section.

³ But for some mechanical systems one can use the energy, for some chemical systems the entropy, and for some genetic systems the mean fitness.

⁴ $(\bar{v}, \bar{p})$ belongs to the ω -limit set of $(v(0), p(0))$ iff there is a sequence $t_n \rightarrow \infty$ such that $(v(t_n), p(t_n)) \rightarrow (\bar{v}, \bar{p})$; clearly points on the same orbit have the same ω -limit set, so we also speak about the ω -limit set of an orbit; if one considers sequences $t_n \rightarrow -\infty$ one speaks about the α -limit set; note that α is the first and ω the last character in the Greek alphabet.

⁵ The appropriate version of Lyapunov's theorem goes by the name of LaSalle's Invariance Principle, see [Krasovskii-LaSalle principle \(Wikipedia\)](#).

We conclude that density dependence in prey population growth is a stabilising mechanism for prey-predator interaction.

SECTION 2.4

Deriving the Holling Type II Functional Response by a Time Scale Argument

The *functional response* is, by definition, the number of prey eaten per predator per unit time, usually expressed as a function of prey density. The *numerical response* is the number of offspring produced per predator per unit time (as a result of eating prey). A frequent assumption (which ignores reproduction delays, etc.) is that the numerical response equals the functional response times a constant, the conversion efficiency.

EXERCISE 2.18. What is the functional response in the Volterra-Lotka model? What parameter combination corresponds to the conversion efficiency?

SOLUTION. In the Volterra-Lotka model, the per-predator consumption rate is bv , so the model uses a Holling type I functional response. Comparing the prey-loss term bvp with the predator-gain term dvp shows that the conversion efficiency is $\frac{d}{b}$. $\square$

As the catching and “handling” of prey takes time and the digestive tract has only limited capacity, the functional response should saturate at high prey density. The so-called Holling type II functional response

$$v \mapsto \frac{bv}{1 + b\beta v} \quad (2.13)$$

does indeed approach a limit $\frac{1}{\beta}$ for $v \rightarrow \infty$. The aim of this section is to derive this expression by a time-scale argument from a more complicated model in which we distinguish two types of predators, those searching for prey and those busy handling a prey caught earlier:

$$\frac{dv}{dt} = av - bvs, \quad (2.14)$$

$$\frac{ds}{dt} = -bvs + \frac{1}{\beta}(1 + \eta)h - cs, \quad (2.15)$$

$$\frac{dh}{dt} = bvs - \frac{1}{\beta}h - ch. \quad (2.16)$$

Here s denotes the searching predators and h the handling predators, so $p = s + h$. Upon catching prey, a predator changes from searching to handling. Handling is completed at rate $\frac{1}{\beta}$, whereupon the predator starts searching again and, in addition, produces $\eta = \frac{d}{b}$ offspring (which start life searching). Equivalently, the handling time is an exponentially distributed random variable with mean β , and the average energy content of one prey item suffices to produce η predators.

EXERCISE 2.19. Assume $\eta, \beta a, \beta c$ and $\frac{p_0}{v_0}$ are all very small and of the same order of magnitude.⁶ Meanwhile, assume that bp_0 is of the same order of magnitude as a .⁷ Perform a two-time-scale

⁶ This means that very little predator biomass can be produced from one prey item, that the handling time is negligible on the time scales of prey population growth and a predator’s expected lifetime, and that there are far fewer predators than prey.

⁷ This means that predation changes prey density on the same time scale as intrinsic prey population growth.

analysis of (2.16) in the spirit of *Chapter 1*. In particular, derive the system

$$\begin{aligned}\frac{dv}{dt} &= av - \frac{bvp}{1 + b\beta v}, \\ \frac{dp}{dt} &= -cp + \frac{dvp}{1 + b\beta v},\end{aligned}\tag{2.17}$$

for changes in the prey and predator densities at the slow time scale.

SOLUTION. bvs represents the rate at which predators change from searching to handling, whereas $\frac{h}{\beta}$ is the rate at which handling is completed and predators return to searching. On the slow time scale, these fast transitions are approximately in equilibrium, so $bvs \approx \frac{1}{\beta}h$. Then $h = bvs\beta$, $p = h + s = bvs\beta + s$, and $s = \frac{p}{bv\beta + 1}$.

$$\frac{dv}{dt} = av - \frac{bvp}{bv\beta + 1}, \quad \frac{dp}{dt} = \frac{d(h + s)}{dt} = \frac{dh}{dt} + \frac{ds}{dt} = -cp + \frac{dvp}{1 + bv\beta}.$$

□

SECTION 2.5

| The Destabilising Effect of a Saturating Functional Response

- EXERCISE 2.20. (i) Draw the isoclines and the wind-direction field for system (2.17), assuming that $\frac{d}{b\beta} > c$.
(ii) Show that the non-trivial steady state of (2.17) is unstable.
(iii) Formulate a conclusion in biological terms (taking the title of this section as a hint).

SOLUTION. (i)

$$\begin{aligned}\frac{dv}{dt} &= av - \frac{bvp}{1+b\beta v} = 0 \implies av = \frac{bvp}{1+b\beta v} \\ \frac{dp}{dt} &= -cp + \frac{dvp}{1+b\beta v} = 0 \implies cp = \frac{dvp}{1+b\beta v}\end{aligned}$$

If $\frac{d}{b\beta} < c$, $v < 0$, it is biologically meaningless.

(ii)

$$\begin{aligned}\frac{dv}{dt} = 0 &\implies av - \frac{bvp}{1+b\beta v} = 0 \implies p = \frac{a(1+b\beta v)}{b} \\ \frac{dp}{dt} = 0 &\implies -cp + \frac{dvp}{1+b\beta v} = 0 \implies v = \frac{c}{d-cb\beta}\end{aligned}$$

Therefore

$$p = \frac{a\left(1 + \frac{b\beta c}{d-cb\beta}\right)}{b} = \frac{a}{b} + \frac{\beta c}{d-cb\beta} = -\frac{ad}{b(-d+bc\beta)}.$$

$$J = \begin{bmatrix} \frac{\partial}{\partial v}\left(av - \frac{bvp}{1+b\beta v}\right) & \frac{\partial}{\partial p}\left(av - \frac{bvp}{1+b\beta v}\right) \\ \frac{\partial}{\partial v}\left(-cp + \frac{dvp}{1+b\beta v}\right) & \frac{\partial}{\partial p}\left(-cp + \frac{dvp}{1+b\beta v}\right) \end{bmatrix}$$

For steady state $\left(\frac{c}{d-cb\beta}, \frac{a}{b} + \frac{c\beta}{d-cb\beta}\right)$:

$$J = \begin{bmatrix} a + \frac{abcd\beta}{(-d+cb\beta)^2\left(1-\frac{bc\beta}{-d+bc\beta}\right)^2} + \frac{ad}{(-d+bc\beta)\left(1-\frac{bc\beta}{-d+bc\beta}\right)} & \frac{bc}{(-d+bc\beta)\left(1-\frac{bc\beta}{-d+bc\beta}\right)} \\ \frac{-acd^2\beta}{(-d+cb\beta)^2\left(1-\frac{bc\beta}{-d+bc\beta}\right)^2} - \frac{ad^2}{b(-d+bc\beta)\left(1-\frac{bc\beta}{-d+bc\beta}\right)} & -c + \frac{cd}{(-d+bc\beta)\left(1-\frac{bc\beta}{-d+bc\beta}\right)} \end{bmatrix}$$

$$\lambda_1 = \frac{abc^2\beta - \sqrt{ac}b\sqrt{-4d^2 + 4bcd\beta + ab^2c\beta^2}}{2bd}, \quad \lambda_2 = \frac{ab^2c\beta + \sqrt{ac}b\sqrt{-4d^2 + 4bcd\beta + ab^2c\beta^2}}{2bd}.$$

It is clear that λ_2 has a positive real part, so the equilibrium is unstable.

- (iii) When $\frac{d}{b\beta} - c > a$, the predator population grows faster than the prey population, eventually causing the prey population to decline. Once the prey population diminishes, the predator population also declines because food is scarce.

When $\frac{d}{b\beta} - c < a$, the predator cannot keep pace with prey growth at high prey density, so the prey escapes predator control. Thus a saturating functional response is destabilising compared with the Volterra–Lotka model.

□

Remark. It is a somewhat delicate task to determine where the orbits of (2.17) ultimately go. When prey density gets very large, the predator density satisfies approximately

$$\frac{dp}{dt} = \left(\frac{d}{b\beta} - c \right) p,$$

so p grows exponentially with exponent $\frac{d}{b\beta} - c$. When $\frac{d}{b\beta} - c < a$, the prey will escape from predator control, in the sense that both grow exponentially but the predator at a slower rate. When, on the other hand, $\frac{d}{b\beta} - c > a$, the predator overtakes the prey and causes its decline. So then cyclic alternations of growth and decline of each of the two species continue indefinitely.

SECTION 2.6

| The Rosenzweig-MacArthur Model

Combining the two modifications of Volterra-Lotka discussed so far, we obtain the system

$$\begin{aligned}\frac{dv}{dt} &= v \left(a - ev - \frac{bp}{1 + b\beta v} \right), \\ \frac{dp}{dt} &= p \left(-c + \frac{dv}{1 + b\beta v} \right).\end{aligned}\tag{2.18}$$

Both this system, and the graphical stability criterion that we shall derive in the next exercise, carry the names of Rosenzweig and MacArthur. The system (2.18) is of the form

$$\begin{aligned}\frac{dv}{dt} &= v(h(v) - p\phi(v)), \\ \frac{dp}{dt} &= p(-c + \psi(v)).\end{aligned}\tag{2.19}$$

with

- (i) h decreasing and $h(K) = 0$ for some $K > 0$;
- (ii) the functional response $v \mapsto v\phi(v)$ is increasing;
- (iii) the numerical response $v \mapsto \psi(v)$ is increasing;

to which we add

- (iv) $\psi^{-1}(c) < K$,

to avoid that the predator is doomed to go extinct. Systems of the form (2.19) are sometimes called Kolmogorov-type prey-predator systems (see Freedman, 1980).

EXERCISE 2.21. (i) Demonstrate that (2.19) has exactly one steady state with both species present.

(ii) Compute the Jacobian matrix at this steady state and show that the determinant is positive.

(iii) Relate the sign of the trace of the Jacobian matrix to the sign of the derivative of $v \mapsto \frac{h(v)}{\phi(v)}$ at the steady-state value of v . In other words, determine whether the prey isocline $p = \frac{h(v)}{\phi(v)}$ is decreasing or increasing at the steady state.

(iv) Formulate carefully how the stability of the steady state can be read off from the isocline picture.

(v) Check the result of (iv) against your findings in Section 3.3, system (2.9), and Section 3.5, system (2.17).

(vi) Use *Poincaré–Bendixson theory* to show that whenever

- the internal steady state is unstable,
- orbits stay bounded (i.e., cannot escape to infinity),

there must exist at least one limit cycle.

(vii) Apply the results of (iv) and (vi) to system (2.18).

(viii) What can you say about Hopf bifurcation for system (2.18)?

SOLUTION. (i)

$$\frac{dv}{dt} = v \left(a - ev - \frac{bp}{1+b\beta v} \right) = v(h(v) - p\phi(v))$$

$$\frac{dp}{dt} = p \left(-c + \frac{dv}{1+b\beta v} \right) = p(-c + \psi(v))$$

Let $\frac{dv}{dt} = 0$

$$v(h(v) - p\phi(v)) = 0 \implies v = 0, \text{ or } h(v) = p\phi(v)$$

Let $\frac{dp}{dt} = 0$

$$p(-c + \psi(v)) = 0 \implies p = 0, \text{ or } \psi(v) = c$$

Thus $h(v) = p\phi(v)$ and $\psi(v) = c$ give the steady state with both species present. For uniqueness, since h is decreasing and $v\phi(v)$ is increasing, the curves of $h(v)$ and $p\phi(v)$ intersect at exactly one point for a given value of p . Thus, for any given p , there is a unique v such that $h(v) = p\phi(v)$. Because $\psi^{-1}(c) < K$, we know that the solution of $\psi(v) = c$ is less than K . Since $h(K) = 0$ and h is decreasing, $h(\psi^{-1}(c)) > 0$. Thus the unique steady state with $p > 0$ also has $v > 0$.

(ii) Let

$$\begin{aligned} \frac{dv}{dt} &= v(h(v) - p\phi(v)) = f(v, p), \\ \frac{dp}{dt} &= p(-c + \psi(v)) = g(v, p). \end{aligned}$$

Then the Jacobian is

$$J = \begin{bmatrix} \frac{\partial f}{\partial v} & \frac{\partial f}{\partial p} \\ \frac{\partial g}{\partial v} & \frac{\partial g}{\partial p} \end{bmatrix} = \begin{bmatrix} h(v) + v \frac{dh(v)}{dv} - p\phi(v) - vp \frac{d\phi(v)}{dv} & -v\phi(v) \\ p \frac{d\psi(v)}{dv} & -c + \psi(v) \end{bmatrix}$$

$$\det(J(v^*, p^*)) = \left(h(v^*) + v^* \frac{dh(v^*)}{dv^*} - p^* \phi(v^*) - v^* p^* \frac{d\phi(v^*)}{dv^*} \right) (-c + \psi(v^*)) + v^* \phi(v^*) p^* \frac{d\psi(v^*)}{dv^*}$$

Since $h(v^*) = p^* \phi(v^*)$, h is a decreasing function and $\phi(v)$ is increasing. It is evident that $v^* \phi(v^*) p^* \frac{d\psi(v^*)}{dv^*} > 0$, $h(v^*) + v^* \frac{dh(v^*)}{dv^*} < 0$ due to h being decreasing. $\psi(v^*) - c = 0$ since $\psi(v^*) = c$. Therefore, $\det(J(v^*, p^*)) > 0$.

(iii)

$$h(v) = a - ev, \quad \phi(v) = \frac{b}{1+b\beta v}, \quad \psi(v) = \frac{dv}{1+b\beta v}, \quad k = \frac{a}{e}$$

Since $\psi^{-1}(c) < k$, we have $\frac{c}{d-b\beta c} < \frac{a}{e}$, which implies $ce < ad - ab\beta c$, thus $ad \geq ce + ab\beta c$.

$$J = \begin{bmatrix} a - ev - \frac{bp}{1+b\beta v} + v \left(-e + \frac{b^2 p \beta}{(1+b\beta v)^2} \right) & -\frac{bv}{1+b\beta v} \\ p \left(-\frac{bv d \beta}{(1+b\beta v)^2} + \frac{d}{1+b\beta v} \right) & -c + \frac{dv}{1+b\beta v} \end{bmatrix}$$

At the steady state $(v^*, p^*) = \left(\frac{c}{d-b\beta c}, p^* \right)$:

$$J(v^*, p^*) = \begin{bmatrix} \frac{-bd^3\beta - 3b^3c^2dp\beta^2 + b^4c^3p\beta^3 + ad^2(d-bc\beta) + cd^2(-2e+3b^2p\beta)}{d^2(d-bc\beta)} & -\frac{bc}{d} \\ \frac{p(d-bc\beta)^2}{d} & 0 \end{bmatrix}$$

$$\text{Tr}(J) = \frac{-bd^3\beta - 3b^3c^2dp\beta^2 + b^4c^3p\beta^3 + ad^2(d-bc\beta) + cd^2(-2e+3b^2p\beta)}{d^2(d-bc\beta)},$$

$\det(J)$ is positive. Therefore $p = \frac{h(v)}{\phi(v)}$ is decreasing at the steady state.

- (iv) The points where the isoclines intersect are the steady states of the system. Each intersection represents a situation in which neither population is changing. On each isocline, indicate the direction in which trajectories move. If trajectories approach the steady state from multiple directions and appear to be attracted to it, the steady state is asymptotically stable. If trajectories are repelled from it, the steady state is unstable. If $p = \frac{h(v)}{\phi(v)}$ is steeper than the predator isocline near the steady state, then trajectories circle the steady state counter-clockwise. If the predator isocline is steeper, then trajectories circle clockwise.
- (v) System 3.3:

$$\begin{cases} \frac{dv}{dt} = v(a - ev - bp) \\ \frac{dp}{dt} = p(-c + dv) \end{cases} \implies \begin{cases} h(v) = a - ev \\ \psi(v) = dv \\ \phi(v) = b \end{cases}$$

- $h(v) = 0 \implies v = \frac{a}{e}$.
- $v\phi(v) = bv$ is increasing.
- $\psi(v) = dv$ is increasing.
- $\psi^{-1}(c) = \frac{c}{d} < \frac{a}{e}$ is true when $-c + d\frac{a}{e} > 0$.

System 3.4.5:

$$\begin{cases} \frac{dv}{dt} = av - \frac{bvp}{1+b\beta v} \\ \frac{dp}{dt} = -cp + \frac{dvp}{1+b\beta v} \end{cases} \implies \begin{cases} h(v) = a \\ \psi(v) = \frac{dv}{1+b\beta v} \\ \phi(v) = \frac{b}{1+b\beta v} \end{cases}$$

Since $h(v) = a$, there is no finite K for which $h(K) = 0$, and $h(v)$ is constant rather than decreasing.

- (vi) Given that orbits stay bounded, and since the internal steady state is unstable, trajectories originating near this steady state will move away. Therefore, in some neighbourhoods of this steady state, we have no equilibria. Now, we need to find a trapping region. A trapping region is a region in the phase space from which trajectories cannot escape. Given that orbits stay bounded, it means trajectories do not go off to infinity. Combining this with the fact that they move away from the unstable steady state, we know that there exists a trapping region, which is a bounded region enclosing the unstable state and from which trajectories do not escape. Since the *Poincaré-Bendixson theory* tells us that if we have a trajectory that stays entirely within a trapping region and there are no equilibrium points within that region, there must be a limit cycle within that region. Given that the trapping region encloses the unstable steady state and trajectories are bounded, we can say that the limit cycle exists.
- (vii) From (iv) and (vi), if

$$\left(\frac{c}{d - bc\beta}, \frac{ad^2 - cde - abcd\beta}{b(d - bc\beta)^2} \right)$$

is unstable, then there must be a limit cycle.

- (viii) If $ad < ce + ab\beta c$, the orbit tends to a limit cycle. If $ad > ce + ab\beta c$, the orbit tends to the internal stable point. Thus, $ad = ce + ab\beta c$ is the critical value at which the system's stability changes as a parameter varies.

□

EXERCISE 2.22. Show that under the scaling $\tau = ct$, $x = b\beta v$, $y = \frac{b}{a}p$, system (2.18) transforms into

$$\begin{aligned} \varepsilon \frac{dx}{d\tau} &= x \left(1 - \frac{x}{K} - \frac{y}{1+x} \right), \\ \frac{dy}{d\tau} &= y \left(-1 + \theta \frac{x}{1+x} \right), \end{aligned} \tag{2.20}$$

where $\varepsilon = \frac{c}{a}$, $\theta = \frac{d}{b\beta c}$, and $K = \frac{b\beta a}{e}$. Assume that $\theta > 1$ and $K > 1 + \frac{2}{\theta-1}$.

Sketch the phase portrait for $\varepsilon \ll 1$ (this is quite subtle!). What does this assumption mean? Summarise your conclusions, i.e., interpret the phase portrait in biological terms.

Hints. The rescaling $\tau = \varepsilon t$ (where t differs from the original t) can be used to show that on the fast time scale t the y -component is approximately constant while x varies according to

$$\frac{dx}{dt} = x \left(1 - \frac{x}{K} - \frac{y}{1+x} \right), \quad (2.21)$$

where y is a parameter. By plotting the curve

$$y = (1+x) \left(1 - \frac{x}{K} \right) \quad (2.22)$$

in the positive (x, y) -quadrant, one can obtain an overview of the steady states of (2.21) and their stability as functions of y .

SOLUTION. Let $\tau = ct$, $x = b\beta v$, $y = \frac{b}{a}p$.

$$\frac{dx}{d\tau} = \frac{dx}{dv} \frac{dv}{dt} \frac{dt}{d\tau} = b\beta \frac{dv}{dt} \frac{1}{c} = \frac{b\beta x}{b\beta c} \frac{1}{c} \left(a - \frac{e}{b\beta} x - \frac{b\frac{a}{b}y}{1+x} \right) = \frac{ax}{c} \left(1 - \frac{e}{b\beta a} x - \frac{y}{1+x} \right)$$

$$\frac{dy}{d\tau} = \frac{dy}{dp} \frac{dp}{dt} \frac{dt}{d\tau} = \frac{b}{a} \frac{dp}{dt} \frac{1}{c} = \frac{b}{a} \left(\frac{a}{b} y \left(-c + \frac{d\frac{x}{b\beta}}{1+x} \right) \right) \frac{1}{c} = y \left(-1 + \frac{dx}{b\beta c(1+x)} \right)$$

Given $\theta = \frac{d}{b\beta c}$,

$$\frac{dy}{d\tau} = y \left(-1 + \frac{\theta x}{1+x} \right)$$

Given $\varepsilon = \frac{c}{a}$, $K = \frac{b\beta a}{e}$,

$$\varepsilon \frac{dx}{d\tau} = x \left(1 - \frac{x}{K} - \frac{y}{1+x} \right)$$

When $\varepsilon \ll 1$, the dynamics of x are much faster than those of y . The x -nullcline is

$$x \left(1 - \frac{x}{K} - \frac{y}{1+x} \right) = 0$$

Biologically, when $\varepsilon \ll 1$, it implies the prey adjusts rapidly and grows much more quickly than predators. So this can lead to oscillatory dynamics where there are a large number of resources for predators, and then the number of predators will increase due to abundant food, and consume more prey, thus the cycle repeats. $\square$

The *slow dynamics* are described by

$$\frac{dy}{d\tau} = y \left(-1 + \theta \frac{x(y)}{1+x(y)} \right), \quad (3.6.6)$$

with $x(y)$ a stable steady state of (2.21).

EXERCISE 2.23. Summarise (in a mixture of mathematical and biological terms) your knowledge about (2.18). You may use the information (which we now provide) that it has been proven that (2.18) admits at most one limit cycle (the proof is far from easy). A convenient way to sketch how the qualitative behaviour depends on the parameters is to draw the diagram as in Figure 3.3, and to describe the dynamical behaviour (in biological terms) for each of the three parameter domains.

SOLUTION. From (2.18), we know that the steady state at which both species are present may be stable or unstable. If it is unstable, there will be a limit cycle.

- For (3), $\frac{a}{c}$ is much larger than $b\beta$, which means that the capacity of prey is large enough, so it is not useful for future development. In this situation, potentially complex dynamics.
- For (2), the capacity of prey and predation efficiency in this prey-predator system will be balanced.
- For (1), the low capacity of prey and high predation efficiency might lead to insufficient food for predators, and then lead to the extinction of the predator.

□

CHAPTER 3

Movement in Space

SECTION 3.1

| Flux

SECTION 3.2

| Various Ways to Motivate Fick's Law

SECTION 3.3

| Transport by Diffusion

SECTION 3.4

How to Measure the Diffusion Coefficient

A capillary tube is inserted into a suspension of bacteria of known concentration. After a prescribed period of time, the tube is extracted and the number of bacteria that have entered is counted. Assume that the bacteria can be described in terms of a concentration u , that they move randomly, that the concentration at the mouth of the tube is always a constant, u_0 say, that there are no bacteria in the tube at the beginning of the experiment, and that the concentration in the tube varies only in the length direction and not in the radial direction. A mathematical formulation of these assumptions reads

$$\frac{\partial u}{\partial t} = d \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < \infty, \quad t > 0, \quad (3.1)$$

$$u(t, 0) = u_0, \quad t > 0, \quad (3.2)$$

$$u(0, x) = 0, \quad x > 0, \quad (3.3)$$

where x measures the distance down the (infinitely long, by debatable assumption) tube.

EXERCISE 3.1. Derive the expression

$$d = \frac{\pi N^2}{4u_0^2 A^2 T}, \quad (3.4)$$

where N denotes the number of bacteria in the tube at time T and A the cross-sectional area of the tube.

SOLUTION. Take as a starting point the *fundamental solution*

$$G(t, x) = \frac{1}{2\sqrt{\pi dt}} e^{-\frac{x^2}{4dt}},$$

which is the solution of the diffusion equation with initial data $u(0, x) = \delta(x)$. If we replace this initial condition by the general condition $u(0, x) = g(x)$, for $x \in \mathbb{R}$, then

$$u(t, x) = \frac{1}{2\sqrt{\pi dt}} \int_{\mathbb{R}} e^{-\frac{(x-y)^2}{4dt}} g(y) dy = \frac{1}{2\sqrt{\pi dt}} \int_{-\infty}^0 2u_0 e^{-\frac{(x-y)^2}{4dt}} dy,$$

where we are given that

$$g(y) = \begin{cases} 0, & y > 0 \\ 2u_0, & y < 0 \end{cases}.$$

As $x \rightarrow 0$,

$$\begin{aligned} u(t, x) &= \frac{u_0}{\sqrt{\pi dt}} \int_{-\infty}^0 e^{-\frac{y^2}{4dt}} dy \\ &= \frac{2u_0 \sqrt{dt}}{\sqrt{\pi dt}} \int_0^{\infty} e^{-\frac{y}{\sqrt{4dt}}} d\left(\frac{y}{\sqrt{4dt}}\right)^2 \\ &= \frac{2u_0}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = u_0. \end{aligned}$$

For $y \leq 0$, we change the variable

$$\xi = \frac{x-y}{2\sqrt{dt}},$$

then

$$dy = -2\sqrt{dt} d\xi, \quad y = -2\sqrt{dt}\xi + x.$$

Let $y = 0$, we have $\xi = \frac{x}{2\sqrt{dt}}$, which is the lower limit, and the upper limit is ∞ . We have

$$u(t, x) = \frac{u_0}{2\sqrt{\pi dt}} \int_{\frac{x}{2\sqrt{dt}}}^{\infty} 4\sqrt{dt} e^{-\xi^2} d\xi = \frac{2u_0}{\sqrt{\pi}} \int_{\frac{x}{2\sqrt{dt}}}^{\infty} e^{-\xi^2} d\xi,$$

i.e.,

$$\frac{u(t, x)}{u_0} = \frac{2}{\sqrt{\pi}} \int_{\frac{x}{2\sqrt{dt}}}^{\infty} e^{-\xi^2} d\xi.$$

Therefore

$$\begin{aligned} \int u dx &= \int_0^{\infty} \frac{2u_0}{\sqrt{\pi}} \int_{\frac{x}{2\sqrt{dt}}}^{\infty} e^{-\xi^2} d\xi dx \\ &= \frac{2u_0}{\sqrt{\pi}} \int_0^{\infty} \int_{\frac{x}{2\sqrt{dt}}}^{\infty} e^{-\xi^2} d\xi dx \\ &= \frac{4\sqrt{dt}u_0}{\sqrt{\pi}} \int_0^{\infty} \int_{\zeta}^{\infty} e^{-\xi^2} d\xi d\zeta \quad (\text{Let } \zeta = \frac{x}{2\sqrt{dt}}) \\ &= \frac{4\sqrt{dt}u_0}{\sqrt{\pi}} \int_0^{\infty} \int_0^{\xi} e^{-\xi^2} d\zeta d\xi \\ &= \frac{4\sqrt{dt}u_0}{\sqrt{\pi}} \int_0^{\infty} \xi e^{-\xi^2} d\xi \\ &= \frac{4\sqrt{dt}u_0}{\sqrt{\pi}} \cdot \frac{1}{2} = \frac{2\sqrt{dt}u_0}{\sqrt{\pi}}. \end{aligned}$$

Since

$$N = A \int u dx = \frac{2\sqrt{dt}u_0}{\sqrt{\pi}} A,$$

we conclude that $d = \frac{\pi N^2}{4u_0^2 A^2 T}$. □

CHAPTER 4

Linear Diffusion

SECTION 4.1

| The Fundamental Solution

SECTION 4.2

| Separation of Variables and Spectral Theory

SECTION 4.3

| The Asymptotic Speed of Propagation

Reaction-Diffusion Equations

| Introduction

SECTION 5.2

| Stability Criteria for Uniform Steady States

SECTION 5.3

Scalar Reaction-Diffusion Equations

Consider once more the zero-flux nonlinear boundary value problem

$$u_{xx} + f(u) = 0, \quad (5.1)$$

$$u_x(0) = 0 = u_x(L), \quad (5.2)$$

with parameter L . We now want to derive relations between solutions by using extension and symmetry arguments instead of phase plane analysis.¹

EXERCISE 5.1. Show that if u is a solution of (5.1)–(5.2), so is w , defined by $w(x) = u(L - x)$. We call u *symmetric* if $w = u$. What symmetry does this amount to?

SOLUTION. By the chain rule,

$$w'(x) = \frac{d}{dx} u(L - x) = -u'(L - x),$$

$$w''(x) = u''(L - x).$$

Hence,

$$w''(x) + f(w(x)) = u''(L - x) + f(u(L - x)) = 0,$$

since u satisfies (5.1) and this identity holds after replacing x by $L - x$. For the boundary conditions,

$$\begin{cases} w_x(0) = -u_x(L) = 0 \\ w_x(L) = -u_x(0) = 0 \end{cases} \quad ,$$

Therefore, $w(x) = u(L - x)$ satisfies both the differential equation and the boundary conditions, and hence w is also a solution. The function u is symmetric about the midpoint of the interval $[0, L]$, namely $\frac{L}{2}$; equivalently,

$$u\left(\frac{L}{2} + s\right) = u\left(\frac{L}{2} - s\right), \quad \forall s.$$

It means that if we fold the graph of u along the vertical line $x = \frac{L}{2}$, the two halves would match up perfectly. This is the reflection symmetry with respect to the line $x = \frac{L}{2}$. $\square$

EXERCISE 5.2. Assume $f(0) = 0$, $f'(0) > 0$. Show that then bifurcations occur if $L = \frac{k\pi}{\sqrt{f'(0)}}$. Consider $k = 1$. Give arguments in favour of the claim that the bifurcating solutions are *not* symmetric. Show that, consequently, the bifurcation must be a pitchfork.

SOLUTION. Given that $f(0) = 0$ and $f'(0) > 0$, a *Taylor expansion* gives

$$f(u) \approx f(0) + f'(0)u \approx f'(0)u,$$

where $f(0) = 0$. Substituting this into our differential equation gives

$$u_{xx} + f'(0)u = 0.$$

Assume the solutions are of the form $u(x) = e^{mx}$, since $f'(0) > 0$,

$$m^2 e^{mx} + f'(0)e^{mx} = 0 \implies m^2 + f'(0) = 0 \implies m = \pm i\sqrt{f'(0)}.$$

¹ The key advantage of such arguments is that they also work for systems.

By Euler's formula, the general solution is

$$u(x) = A \cos(\sqrt{f'(0)}x) + B \sin(\sqrt{f'(0)}x).$$

Given that $u_x(0) = 0$, $B\sqrt{f'(0)} = 0$, then $B = 0$, and

$$u_x(L) = -A\sqrt{f'(0)} \sin(\sqrt{f'(0)}L) = 0.$$

For non-trivial solution,

$$\sin(\sqrt{f'(0)}L) = 0 \iff \sqrt{f'(0)}L = k\pi,$$

where k is an integer. Hence

$$L = \frac{k\pi}{\sqrt{f'(0)}}.$$

A solution $u(x)$ on $[0, L]$ that satisfies the given boundary-value problem can be extended to $[-L, L]$ by even reflection. On this extended domain, we only need the boundary condition at $x = L$, which is $u_x(L) = 0$. By even reflection, we also have $u_x(-L) = 0$. The first non-trivial solution on this extended domain will occur when the solution completes half a period by $x = L$, which is $L = \frac{k\pi}{\sqrt{f'(0)}}$. Consider

$k = 1$, $L = \frac{\pi}{\sqrt{f'(0)}}$. Since $B = 0$, we have $u(x) = A \cos(\sqrt{f'(0)}x)$, hence

$$u_x\left(\frac{L}{2}\right) = -A\sqrt{f'(0)} \sin\left(\sqrt{f'(0)}\frac{L}{2}\right) = -A\sqrt{f'(0)} \sin\left(\frac{\pi}{2}\right) \neq 0.$$

Therefore a symmetric solution cannot be the bifurcating one, and the bifurcating solutions are not symmetric about the midpoint.

Since the problem is invariant under the reflection $\mathcal{R}u(x) := u(L - x)$ and non-symmetric solutions occur, we established that the bifurcating solutions are not symmetric about the midpoint. Hence such solutions must occur in mirror pairs $\{u, \mathcal{R}u\}$. This loss of symmetry from the trivial branch $u \equiv 0$ at $L = \frac{\pi}{\sqrt{f'(0)}}$ indicates the existence of a pitchfork. $\square$

EXERCISE 5.3. Whenever u is a solution, extend it to a $2L$ -periodic function by

$$\begin{aligned} u(-x) &:= u(x), & 0 \leq x \leq L, \\ u(x + 2L) &:= u(x). \end{aligned}$$

Show that the extension is a solution for parameter value kL , $k = 1, 2, 3, \dots$. Conclude that the branch bifurcating for $k = 1$ repeats itself for every higher value of k .

SOLUTION. The extension $u(x + 2L) = u(x)$ makes the function periodic with a period of $2L$. The extension $u(-x) := u(x)$ reflects the function over the y -axis. Thus, for any given x in $[0, L]$, u has the same value at $-x$. So we have $u_x(-x) = -u_x(x)$. Since $u_x(0) = 0$, the even extension is C^1 at $x = 0$. Moreover, since u solves (5.1) on both sides, u_{xx} matches at $x = 0$ as well. So the extension is $C^2([-L, L])$. Now, extend by periodicity: $u_x(x + 2L) = u_x(x)$. Then

$$u_x(x + 2L) = u_x(x), \quad u_{xx}(x + 2L) = u_{xx}(x).$$

Substitute into the differential equation,

$$u_{xx}(x) + f(u(x)) = 0 \implies u_{xx}(x + 2L) + f(u(x + 2L)) = 0.$$

Given that $u(x + 2L) = u(x)$, the above equation is true. Hence the $2L$ -periodic extension is a classical solution on $\mathbb{R}$. Because the extension is even about $x = 0$ and $2L$ -periodic, it is also even about $x = L$:

$$u(L + x) = u((L + x) - 2L) = u(x - L) = u(-(x - L)) = u(L - x),$$

so $u_x(L) = 0$. By periodicity, $u_x(nL) = 0$ for all integers n . Therefore, restricting the extended solution to $[0, kL]$ yields (5.1) and $u_x(0) = u_x(kL) = 0$, i.e., it solves the Neumann problem for parameter value kL , $k = 1, 2, 3, \dots$.

$u(x + 2L) = u(x)$ means that the behaviour of the solution over an interval of length $2L$ will repeat itself over every subsequent interval of the same length. The evenness about each nL ensures the Neumann conditions at the endpoints 0 and nL . Furthermore, the reflection ensures the repeatability of the solution's behaviour over the extended interval $[-L, L]$, i.e., $k = 1$. To put it in terms of bifurcation theory, if a bifurcation occurs at $k = 1$, the periodic and reflection properties of the extended solution ensure that the same bifurcation scenario will repeat for higher values of k . Thus the branch bifurcating for $k = 1$ repeats for every higher integer k . $\square$

EXERCISE 5.4. Show that u is symmetric if and only if the extension has period L . Show that all branches corresponding to even k consist of symmetric solutions. How are in that case the two solutions (one for each subbranch) related to each other?

SOLUTION. Define the even extension of $u(x)$ to $(-L, L)$ by

$$v(x) = \begin{cases} u(x), & 0 \leq x \leq L \\ u(-x), & -L \leq x < 0 \end{cases}$$

where v is continuous on $(-L, L)$. We have

$$v(\pm L) = u(L), \quad v(0) = u(0), \quad v'(\pm L) = \mp u'(L) = 0, \quad v'(0) = u'(0) = 0,$$

so $v \in C^2(-L, L)$. On $(0, L)$,

$$v_{xx} + f(v) = u_{xx} + f(u) = 0;$$

on $(-L, 0)$,

$$v_x(x) = -u_x(-x), \quad v_{xx}(x) = u_{xx}(-x),$$

hence

$$v_{xx} + f(v(x)) = u_{xx}(-x) + f(u(-x)) = u_{xx}(-x) + f(u) = 0$$

there as well. Thus v satisfies the ODE on $(-L, L)$. Now define the extension $U : \mathbb{R} \rightarrow \mathbb{R}$ by making v $2L$ -periodic:

$$\begin{aligned} U(x) &= v(x), & x \in (-L, L), \\ U(x + 2L) &= U(x), & \forall x. \end{aligned}$$

Because $v'(\pm L) = 0$ and the equation is autonomous, this periodic pasting keeps $U \in C^2(\mathbb{R})$ and ensures that it satisfies (5.1).

" $\implies$ " Suppose u is symmetric about $x = \frac{L}{2}$, i.e., $u(L - x) = u(x)$, $\forall x \in [0, L]$. Fix any $x \in \mathbb{R}$. Using the $2L$ -periodicity, it suffices to check $x \in [-L, L]$. Then

$$U(x + L) = U(x - L).$$

If $x \in (0, L]$, the right-hand side equals $U(L - x)$. By construction on $(-L, L]$, $U(L - x) = u(L - x) = u(x) = U(x)$ by symmetry. If $x \in (-L, 0]$, then $x + L \in (0, L]$ and a similar reduction gives the same conclusion. Hence

$$U(x + L) = U(L - (x + L)) = U(-x) = U(x)$$

for all x , i.e., the extension has period L .

" $\impliedby$ " Conversely, if the even extension is L -periodic, then for $x \in [0, L]$,

$$u(x) = v(x) = v(x - L) = u(L - x).$$

So u is symmetric about $\frac{L}{2}$.

In conclusion, u is symmetric if and only if its even extension is L -periodic.

Considering the problem in a Sturm–Liouville context:

$$\begin{cases} u_{xx} + k^2 u = 0 \\ u_x(0) = 0 \\ u_x(L) = 0 \end{cases}$$

The general solution to $u_{xx} + k^2 u = 0$ is

$$u(x) = A \cos(kx) + B \sin(kx).$$

For $u_x(0) = 0$,

$$u_x(x) = -Ak \sin(kx) + Bk \cos(kx),$$

then $u_x(0) = Bk = 0$. Since $k \neq 0$, $B = 0$, and $u(x) = A \cos(kx)$. For $u_x(L) = 0$,

$$u_x(L) = -Ak \sin(kL) = 0.$$

For non-trivial solutions where $A \neq 0$,

$$\sin(kL) = 0 \iff k = \frac{n\pi}{L},$$

where n is an integer. Now, if k is even, then n is even, $u(x) = A \cos(kx)$ is symmetric about $x = \frac{L}{2}$ when n is even. This is because the cosine function is even and retains its value when the argument is negated: $\cos(-kx) = \cos(kx)$. Thus, if we have $u(x) = u(L - x)$, this shows symmetry about $x = \frac{L}{2}$.

Nonconstant Neumann solutions on $[0, L]$ arise by taking a segment of a periodic orbit whose minimal period is T with $L = \frac{k}{2}T$. If k is even (write $k = 2m$), then the $2L$ -periodic extension actually has period $L = mT$. Hence the periodic extension U is L -periodic. If U_1 is the periodic extension of one subbranch, then the other subbranch is obtained by a half-period phase shift, and both restrictions to $[0, L]$ satisfy the Neumann conditions and are symmetric about $x = \frac{L}{2}$. The two subbranches are related by

$$U_2(x) = U_1\left(x + \frac{T}{2}\right) = U_1\left(x + \frac{L}{k}\right),$$

and we have

$$U_2'(0) = U_1'\left(\frac{T}{2}\right) = 0, \quad U_2'(L) = U_1'\left(L + \frac{T}{2}\right) = U_1'\left(\frac{T}{2}\right) = 0.$$

On $[0, L]$ this means one solution has a maximum at the endpoints $x = 0, L$ and a minimum in the interior, while the other has a minimum at the endpoints and a maximum in the interior (both remain symmetric). In general, even k give symmetric branches, whereas for odd k the two solutions are related by reflection: $u_+(x) = u_-(L - x)$. $\square$

SECTION 5.4

| The Non-Existence of Patterns for Scalar Equations

SECTION 5.5

| Travelling Waves for Reaction-Diffusion

SECTION 5.6

| Pattern Formation: The Turing Instability

- EXERCISE 5.5. (i) Find the uniform (in x) steady (in t) state.
(ii) Compute the Jacobian matrix of the reaction part.
(iii) Show that the constant steady state is *stable* with respect to homogeneous (i.e., x -independent) perturbations, provided the parameter inequality

$$\frac{R-1}{R+1} < Q \quad (5.3)$$

holds.

SOLUTION. We consider

$$\begin{aligned} \frac{\partial A}{\partial t} &= 1 + R \frac{A^2}{H} - A + \frac{\partial^2 A}{\partial x^2}, \\ \frac{\partial H}{\partial t} &= Q(A^2 - H) + P \frac{\partial^2 H}{\partial x^2}, \end{aligned}$$

with $Q > 0$.

- (i) For the uniform steady state, we set $\frac{\partial A}{\partial t} = 0$ and $\frac{\partial^2 A}{\partial x^2} = 0$, which gives

$$0 = 1 + R \frac{A^2}{H} - A, \quad 0 = Q(A^2 - H).$$

Since $Q > 0$, we have $A^2 = H$, therefore $0 = 1 + R - A$, which implies $A = 1 + R$. Hence the uniform steady state is

$$A = 1 + R, \quad H = (1 + R)^2.$$

- (ii) For

$$f(A, H) = 1 + R \frac{A^2}{H} - A, \quad g(A, H) = Q(A^2 - H),$$

the Jacobian is

$$J = \begin{bmatrix} \frac{\partial f}{\partial A} & \frac{\partial f}{\partial H} \\ \frac{\partial g}{\partial A} & \frac{\partial g}{\partial H} \end{bmatrix} = \begin{bmatrix} \frac{2RA}{H} - 1 & -\frac{RA^2}{H^2} \\ 2QA & -Q \end{bmatrix}.$$

- (iii) Substitute the constant steady state into the Jacobian matrix, giving

$$J = \begin{bmatrix} \frac{R-1}{R+1} & -\frac{R}{(1+R)^2} \\ 2Q(1+R) & -Q \end{bmatrix}.$$

Let

$$\det(J - \lambda I) = \begin{vmatrix} \frac{R-1}{R+1} - \lambda & -\frac{R}{(1+R)^2} \\ 2Q(1+R) & -Q - \lambda \end{vmatrix} = 0,$$

then

$$\text{tr}(J) = \frac{R-1}{R+1} - Q,$$

$$\begin{aligned} \det(J) &= \frac{R-1}{R+1}(-Q) - \left(-\frac{R}{(1+R)^2}\right)(2Q(1+R)) \\ &= -Q \frac{R-1}{R+1} + 2Q \frac{R}{R+1} = Q. \end{aligned}$$

Because $Q > 0$, $\det(J) > 0$ automatically, and $\text{tr}(J) < 0$ holds precisely when $\frac{R-1}{R+1} < Q$. Therefore the uniform steady state is stable with respect to homogeneous perturbations. $\square$

EXERCISE 5.6. (i) Show that the k -th mode, characterised by dependence on x of the form $\cos\left(\frac{k\pi x}{L}\right)$, is stable when

$$P\left(\frac{k\pi}{L}\right)^4 + \left(Q - P\frac{R-1}{R+1}\right)\left(\frac{k\pi}{L}\right)^2 + Q > 0 \quad (5.4)$$

but unstable when the reverse inequality holds.

(ii) Deduce from (5.4) that a necessary condition for instability is

$$Q < P\frac{R-1}{R+1} \quad (5.5)$$

(iii) Deduce, by comparing (5.3) and (5.5), that a necessary condition for pattern-forming instability is $P > 1$ and interpret this condition.

SOLUTION. (i) Given that

$$[Df(\bar{u}) - \mu^2 d]\bar{v} = \lambda \bar{v},$$

$$\lambda^2 + \lambda(-m_{22} - m_{11} + d_1\mu^2 + d_2\mu^2) + ((m_{11} - d_1\mu^2)(m_{22} - d_2\mu^2) - m_{12}m_{21}) = 0.$$

Consider a perturbation around the steady state $A = 1 + R$ and $H = (1 + R)^2$. Let $\bar{v} = (a, h)^\top$, where a and h are perturbations in A and H respectively. Then

$$Df(\bar{u}) = \begin{bmatrix} -1 + 2R\frac{A_0}{H_0} & -R\frac{A_0^2}{H_0^2} \\ 2QA_0 & -Q \end{bmatrix}, \quad d = \text{diag}(1, P).$$

With $\mu = \frac{k\pi}{L}$, one has:

$$m_{11} = -1 + 2R\frac{1+R}{(1+R)^2} = \frac{R-1}{R+1}, \quad m_{12} = -R\frac{(1+R)^2}{(1+R)^4} = -\frac{R}{(1+R)^2},$$

$$m_{21} = 2Q(1+R), \quad m_{22} = -Q, \quad d_1 = 1, \quad d_2 = P.$$

The eigenvalue problem gives the characteristic polynomial

$$\begin{aligned} \lambda^2 + \lambda\left(Q - \frac{R-1}{R+1} + \mu^2 P + \mu^2\right) + \left(\frac{R-1}{R+1} - \mu^2\right)(-Q - P\mu^2) - \left(-\frac{R}{(1+R)^2}\right)2Q(1+R) &= 0 \\ \implies \lambda^2 + \lambda\left(Q - \frac{R-1}{R+1} + \mu^2(P+1)\right) + \left(Q\frac{R-1}{R+1} - Q\mu^2 - P\mu^4 + \frac{R-1}{R+1}P\mu^2 + \frac{2QR}{1+R}\right) &= 0 \\ \implies \lambda^2 + \lambda\left(\underbrace{Q - \frac{R-1}{R+1} + \mu^2(P+1)}_{:=b}\right) + \underbrace{P\mu^4 + \left(Q - P\frac{R-1}{R+1}\right)\mu^2 + Q}_{:=c} &= 0. \end{aligned}$$

For stability, the real parts of λ must be negative. Since

$$\begin{aligned} \begin{cases} \Delta < 0 \\ -\frac{b}{2} < 0 \end{cases} &\implies \begin{cases} c > \frac{b^2}{4} \\ b > 0 \end{cases}, \\ \begin{cases} \Delta > 0 \\ \frac{-b+\sqrt{\Delta}}{2} < 0 \end{cases} &\implies \begin{cases} 0 < c < \frac{b^2}{4} \\ \frac{-b+\sqrt{b^2-4c}}{2} < 0 \end{cases}. \end{aligned}$$

The state is stable, and it will be the case if

$$P\mu^4 + \left(Q - P\frac{R-1}{R+1}\right)\mu^2 + Q > 0.$$

(ii) Given that

$$P\mu^4 + \left(Q - P\frac{R-1}{R+1}\right)\mu^2 + Q = 0,$$

the roots of μ^2 are

$$\mu^2 = \frac{-(Q - P\frac{R-1}{R+1}) \pm \sqrt{(Q - P\frac{R-1}{R+1})^2 - 4PQ}}{2P}.$$

For $\Delta \geq 0$, $(Q - P\frac{R-1}{R+1})^2 \geq 4PQ$. For instability, a positive root is required, so $Q < P\frac{R-1}{R+1}$.

(iii) Pattern-forming instability requires instability to x -dependent perturbations while the homogeneous state remains stable. Homogeneous stability gives (5.3), whereas instability of some spatial mode requires (5.5). Then

$$\frac{R-1}{R+1} < Q < P\frac{R-1}{R+1},$$

hence

$$P > \frac{Q(R+1)}{R-1},$$

which implies $P > 1$ since $Q > \frac{R-1}{R+1}$. This means that the diffusion rate of the inhibitor H is larger than the diffusion rate of the activator A . Under suitable parameter values, this can lead to diffusion-driven instability and the emergence of non-uniform patterns. $\square$

EXERCISE 5.7. Show that the quadratic polynomial in α ,

$$P\alpha^2 + \left(Q - P\frac{R-1}{R+1}\right)\alpha + Q$$

has minimum value

$$-\frac{1}{4}P\left(\frac{R-1}{R+1} - \frac{Q}{P}\right)^2 + Q$$

and check that it is attained for a *positive* value of α when (5.5) holds.

SOLUTION. Substituting

$$\alpha = -\frac{b}{2a} = -\frac{Q - P\frac{R-1}{R+1}}{2P}$$

back to the quadratic polynomial gives

$$\begin{aligned} & P\left(-\frac{Q - P\frac{R-1}{R+1}}{2P}\right)^2 + \left(Q - P\frac{R-1}{R+1}\right)\left(-\frac{Q - P\frac{R-1}{R+1}}{2P}\right) + Q \\ &= \frac{1}{4P}\left(Q - P\frac{R-1}{R+1}\right)^2 - \frac{1}{2P}\left(Q - P\frac{R-1}{R+1}\right)^2 + Q = -\frac{1}{4}P\left(\frac{R-1}{R+1} - \frac{Q}{P}\right)^2 + Q. \end{aligned}$$

If $Q < P\frac{R-1}{R+1}$, we have

$$\frac{R-1}{R+1} - \frac{Q}{P} > 0 \implies -\frac{Q - P\frac{R-1}{R+1}}{2P} > 0,$$

hence $\alpha > 0$. Therefore, the minimum value of the quadratic polynomial is indeed attained for a positive value of α under the given condition. $\square$

EXERCISE 5.8. (Re)formulate the results in biological terms and draw conclusions. In particular, think about how diffusion facilitates pattern formation, what kind of activator-inhibitor system this is, and which modes are potentially unstable for given parameter values.

SOLUTION. The mathematical analysis above shows that diffusion can drive an initially uniform state toward spatial patterning via a diffusion-driven instability. In particular, when the inhibitor diffuses sufficiently faster than the activator ($P > 1$), it leads to the formation of spatial patterns. Biologically, reaction-diffusion can explain how spots and stripes can form on animal skins or how limbs are positioned on the body. The activator-inhibitor system described above states that the activator promotes its own production while also stimulating the production of the inhibitor. The inhibitor suppresses the production of the activator, which in turn suppresses the production of the inhibitor. Such systems are common in nature. For instance, a certain chemical activator promotes its own production but also stimulates the production of an inhibitor. This sets up a feedback loop that can lead to patterns that slow down the production of the activator. For a mode to be unstable, the quadratic polynomial derived from the system must evaluate to a negative value. In nature, we might observe patterns of a particular size or spacing in one situation, and completely different patterns in another, even if the underlying mechanisms are similar. That's because, depending on the parameter values, only certain spatial modes will manifest as patterns. $\square$

CHAPTER 6

Chemotaxis

SECTION 6.1

| Introduction

SECTION 6.2

| Initiation of Pattern Formation

SECTION 6.3

| Derivation of Chemotaxis Models

Infectious Diseases

SECTION 7.1

Infectious Diseases with Demographic Turnover Taken into Account

We consider the following variables

- t time;
- S number of susceptibles;
- i incidence, i.e., the number of new cases per unit time;
- F force of infection, i.e., the probability per unit time that a susceptible is infected;
- B population birth rate;
- μ per capita death rate.

We assume that the infection leads to permanent immunity and that the infection does *not* lead to a higher death rate. These assumptions lead to the following equation

$$\frac{dS}{dt}(t) = B - \mu S(t) - i(t), \quad (7.1)$$

$$i(t) = F(t)S(t), \quad (7.2)$$

$$F(t) = \int_0^\infty i(t - \tau)A(\tau)e^{-\mu\tau}d\tau. \quad (7.3)$$

EXERCISE 7.1. Interpret equation (7.3). In particular, describe in words the meaning of $A(\tau)$.

SOLUTION. (7.3) describes the force of infection, which is the probability per unit time that a susceptible individual will be infected. It is determined by integrating over all possible times in the past, the incidence of new cases at a previous time $t - \tau$, modified by $A(\tau)$ and the exponential decay function $e^{-\mu\tau}$. The function $A(\tau)$ describes the rate at which infectious individuals contact susceptibles and potentially transmit the infection. It is a measure of the interaction between infected and susceptible individuals that could result in transmission, which could be the contact rate function. $\square$

EXERCISE 7.2. Show that the definition

$$S(t) = B \int_0^\infty e^{-\mu a - \int_0^a F(t-a+\sigma) d\sigma} da \quad (7.4)$$

is compatible with (7.1)–(7.2), i.e., show that S defined by (7.4) satisfies (7.1) with $i(t)$ defined by (7.2).

SOLUTION. Rewrite (7.4) as

$$S(t) = B \int_{-\infty}^t e^{-\mu(t-\eta) - \int_0^{t-\eta} F(\eta+\sigma) d\sigma} d\eta = B \int_{-\infty}^t e^{-\mu(t-\eta) - \int_\eta^t F(\sigma) d\sigma} d\eta.$$

Using the *Leibniz integral rule*,

$$\frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t) dx = f(b(t), t) b'(t) - f(a(t), t) a'(t) + \int_{a(t)}^{b(t)} \frac{\partial}{\partial t} f(x, t) dx,$$

we compute

$$\begin{aligned} \frac{dS(t)}{dt} &= B \cdot \frac{d}{dt} \int_{-\infty}^t e^{-\mu(t-\eta) - \int_\eta^t F(\sigma) d\sigma} d\eta \\ &= B \left(e^{-\mu(t-t) - \int_t^t F(\sigma) d\sigma} - 0 + \int_{-\infty}^t \frac{\partial}{\partial t} e^{-\mu(t-\eta) - \int_\eta^t F(\sigma) d\sigma} d\eta \right) \\ &= B \left(1 + \int_{-\infty}^t \frac{\partial}{\partial t} e^{-\mu(t-\eta) - \int_\eta^t F(\sigma) d\sigma} d\eta \right) \\ &= B \left(1 + \int_{-\infty}^t e^{-\mu(t-\eta) - \int_\eta^t F(\sigma) d\sigma} (-\mu - F(t)) d\eta \right) \\ &= B \left(1 - \int_{-\infty}^t (\mu + F(t)) e^{-\mu(t-\eta) - \int_\eta^t F(\sigma) d\sigma} d\eta \right) \\ &= B - \mu B \int_{-\infty}^t e^{-\mu(t-\eta) - \int_\eta^t F(\sigma) d\sigma} d\eta - BF(t) \int_{-\infty}^t e^{-\mu(t-\eta) - \int_\eta^t F(\sigma) d\sigma} d\eta \\ &= B - \mu S(t) - F(t) S(t) = B - \mu S(t) - i(t). \end{aligned}$$

□

EXERCISE 7.3. Describe the meaning of (7.4) when a is interpreted as the age of an individual.

SOLUTION. – $S(t)$: number of susceptibles at time t ;

- B : population birth rate;
- $e^{-\mu a}$: the probability of surviving to age a ;
- $\int_0^a F(t-a+\sigma) d\sigma$: the cumulative force of infection experienced from birth to age a ;
- $\int_0^\infty e^{-\mu a - \int_0^a F(t-a+\sigma) d\sigma} da$: the expected number of remaining individuals who are still susceptible at different ages;
- (7.4): the total number of susceptible individuals in the population at time t .

□

EXERCISE 7.4. By substituting (7.4) into (7.2) and next (7.2) into (7.3), we obtain the non-linear renewal equation

$$F(t) = \int_0^\infty F(t-\tau)B \int_0^\infty e^{-\mu a - \int_0^a F(t-\tau-a+\sigma)d\sigma} da A(\tau) e^{-\mu\tau} d\tau. \quad (7.5)$$

Determine the steady states of this equation.

SOLUTION. Let $F(t) := F$ be a constant. Then

$$\begin{aligned} F &= \int_0^\infty FB \int_0^\infty e^{-\mu a - \int_0^a F d\sigma} da A(\tau) e^{-\mu\tau} d\tau \\ &= FB \int_0^\infty e^{-\mu a - aF} da \int_0^\infty A(\tau) e^{-\mu\tau} d\tau = FB \frac{1}{\mu + F} \int_0^\infty A(\tau) e^{-\mu\tau} d\tau. \end{aligned}$$

Either $F = 0$, which is the disease-free equilibrium, or $F > 0$, in which case we may divide by F . Therefore,

$$1 = B \frac{1}{\mu + F} \int_0^\infty A(\tau) e^{-\mu\tau} d\tau \implies F^* = B \int_0^\infty A(\tau) e^{-\mu\tau} d\tau - \mu.$$

□

EXERCISE 7.5. What is the condition for the (in)stability of the trivial steady state $F = 0$? What is the ecological interpretation of this condition? How does this condition relate to the (non-)existence of a biologically meaningful non-trivial steady state?

SOLUTION. Let $F = 0$, then (7.5) becomes

$$F(t) = \int_0^\infty 0 \cdot B \int_0^\infty e^{-\mu a - \int_0^a 0 d\sigma} da A(\tau) e^{-\mu\tau} d\tau.$$

Consider a small perturbation $F(t) = \xi(t) \neq 0$, where $\xi(t)$ is small. Then

$$\xi(t) = \int_0^\infty \xi(t-\tau)B \int_0^\infty e^{-\mu a} da A(\tau) e^{-\mu\tau} d\tau = \xi(t-\tau)BC \int_0^\infty A(\tau) e^{-\mu\tau} d\tau,$$

where $C = \int_0^\infty e^{-\mu a} da = \frac{1}{\mu}$.

- If $\frac{B}{\mu} \int_0^\infty A(\tau) e^{-\mu\tau} d\tau < 1$, then the perturbation $\xi(t)$ tends to zero, indicating stability.
- If $\frac{B}{\mu} \int_0^\infty A(\tau) e^{-\mu\tau} d\tau > 1$, the perturbation $\xi(t)$ grows, indicating instability.

$F = 0$ is a disease-free state where there is no transmission of infection within the population.

- For $\frac{B}{\mu} \int_0^\infty A(\tau) e^{-\mu\tau} d\tau < 1$, the infection cannot sustain itself in the population. Over time, it will die out because an infected individual does not generate enough new infections before recovery or death. Therefore, there is no biologically meaningful non-trivial steady state for disease transmission, and the disease-free state is stable.
- For $\frac{B}{\mu} \int_0^\infty A(\tau) e^{-\mu\tau} d\tau > 1$, it implies the infection will spread through the population, because each infected individual leads to more than one new case, and it may cause an epidemic. The trivial steady state becomes unstable because any small number of infections will tend to increase, leading to disease persisting in the population. A biologically meaningful endemic steady state can exist.

□

EXERCISE 7.6. Linearise the renewal equation (7.5) around the non-trivial steady state $\bar{F}$.

SOLUTION. Substituting $F(t) = \bar{F} + \xi(t)$ into (7.5), we have

$$\bar{F} + \xi(t) = \int_0^\infty (\bar{F} + \xi(t - \tau)) B \int_0^\infty e^{-\mu a - \int_0^a (\bar{F} + \xi(t - \tau - a + \sigma)) d\sigma} dA(\tau) e^{-\mu \tau} d\tau.$$

Using the first-order expansion

$$e^{-\mu a - \int_0^a (\bar{F} + \xi(t - \tau - a + \sigma)) d\sigma} \approx e^{-\mu a - \bar{F} a} \left(1 - \int_0^a \xi(t - \tau - a + \sigma) d\sigma \right),$$

we have

$$\bar{F} + \xi(t) = \int_0^\infty (\bar{F} + \xi(t - \tau)) B \int_0^\infty e^{-(\mu + \bar{F})a} \left(1 - \int_0^a \xi(t - \tau - a + \sigma) d\sigma \right) dA(\tau) e^{-\mu \tau} d\tau.$$

Therefore,

$$\xi(t) = \int_0^\infty \xi(t - \tau) B \int_0^\infty e^{-\mu a - \bar{F} a} dA(\tau) e^{-\mu \tau} d\tau = \int_0^\infty \xi(t - \tau) B e^{-\mu \tau} A(\tau) d\tau,$$

where $e^\xi \approx 1 + \xi$. □

EXERCISE 7.7. Derive the corresponding characteristic equation

$$1 = \frac{\lambda + \mu}{\lambda + B\bar{A}(\mu)} \frac{\bar{A}(\lambda + \mu)}{\bar{A}(\mu)}, \quad (7.6)$$

where

$$\bar{A}(s) := \int_0^\infty e^{-st} A(t) dt$$

is the Laplace transform of $A(t)$.

SOLUTION. We rewrite (7.6) as

$$\lambda \bar{A}(\mu) + B \bar{A}(\mu)^2 = (\lambda + \mu) \bar{A}(\lambda + \mu).$$

So

$$\lambda \int_0^\infty e^{-\mu \tau} A(\tau) d\tau + B \int_0^\infty e^{-\mu \tau} A(\tau) d\tau \int_0^\infty e^{-\mu \tau} A(\tau) d\tau = (\lambda + \mu) \int_0^\infty e^{-(\lambda + \mu) \tau} A(\tau) d\tau.$$

Given that $\xi(t) = \int_0^\infty \xi(t - \tau) B e^{-\mu \tau} A(\tau) d\tau$, then we have

$$\bar{\xi}(s) = C \bar{\xi}(s) \bar{A}(s + \mu) \implies 1 = C \bar{A}(s),$$

where C is a constant. Since $\bar{A}(\lambda + \mu) = \bar{A}(\mu)$, we have

$$\begin{aligned} & \lambda \int_0^\infty e^{-\mu \tau} A(\tau) d\tau + C \int_0^\infty e^{-\mu \tau} A(\tau) d\tau \int_0^\infty e^{-\mu \tau} A(\tau) d\tau = (\lambda + \mu) \int_0^\infty e^{-\mu \tau} A(\tau) d\tau \\ \implies & C \int_0^\infty e^{-\mu \tau} A(\tau) d\tau \int_0^\infty e^{-\mu \tau} A(\tau) d\tau = \mu \int_0^\infty e^{-\mu \tau} A(\tau) d\tau \implies \frac{C}{\mu} \int_0^\infty A(\tau) e^{-\mu \tau} d\tau = 1. \end{aligned}$$

Hence, when $F = 0$, it is the trivial steady state, and the equation holds. □

COROLLARY 7.1.

The endemic steady state is locally asymptotically stable whenever it exists.

EXERCISE 7.8. As an encore, consider the special case $A(\tau) = \beta e^{-\alpha\tau}$. Verify that in that case (7.2) and (7.3) can be replaced by the familiar ODE

$$\frac{dI}{dt} = \beta IS - (\alpha + \mu)I, \quad (7.7)$$

where $I = \frac{1}{\beta}F$ represents the number of infectious individuals.

SOLUTION. In this special case,

$$F(t) = \int_0^\infty i(t-\tau)\beta e^{-\alpha\tau}e^{-\mu\tau}d\tau = \int_0^\infty F(t-\tau)S(t-\tau)\beta e^{-(\alpha+\mu)\tau}d\tau.$$

Then

$$\begin{aligned} \frac{dF(t)}{dt} &= \frac{d}{dt} \left(\beta \int_0^\infty F(t-\tau)S(t-\tau)e^{-(\alpha+\mu)\tau}d\tau \right) \\ &= \beta \int_0^\infty \frac{\partial}{\partial t} F(t-\tau)S(t-\tau)e^{-(\alpha+\mu)\tau}d\tau \\ &= \beta \int_0^\infty e^{-(\alpha+\mu)\tau} \left(-\frac{\partial}{\partial \tau} F(t-\tau)S(t-\tau) \right) d\tau \\ &= \beta \left(-e^{-(\alpha+\mu)\tau} F(t-\tau)S(t-\tau) \Big|_0^\infty - \int_0^\infty F(t-\tau)S(t-\tau)(\alpha+\mu)e^{-(\alpha+\mu)\tau}d\tau \right) \\ &= \beta F(t)S(t) - (\alpha+\mu)F(t). \end{aligned}$$

Given that $I = \frac{1}{\beta}F$, i.e., $F = \beta I$, we have

$$\beta \frac{dI}{dt} = \beta \beta IS - (\alpha + \mu)\beta I \implies \frac{dI}{dt} = \beta IS - (\alpha + \mu)I,$$

which is exactly the desired ODE. $\square$

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